INCH-POUND
MIL-PRF-8805/8H
6 December 2012
SUPERSEDING
MIL-PRF-8805/8G
10 April 2006

PERFORMANCE SPECIFICATION SHEET

SWITCHES, SENSITIVE, SPDT, LEAF ACTUATOR, LOW LEVEL TO 4 AMPERES, HERMETIC SEAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-PRF-8805.

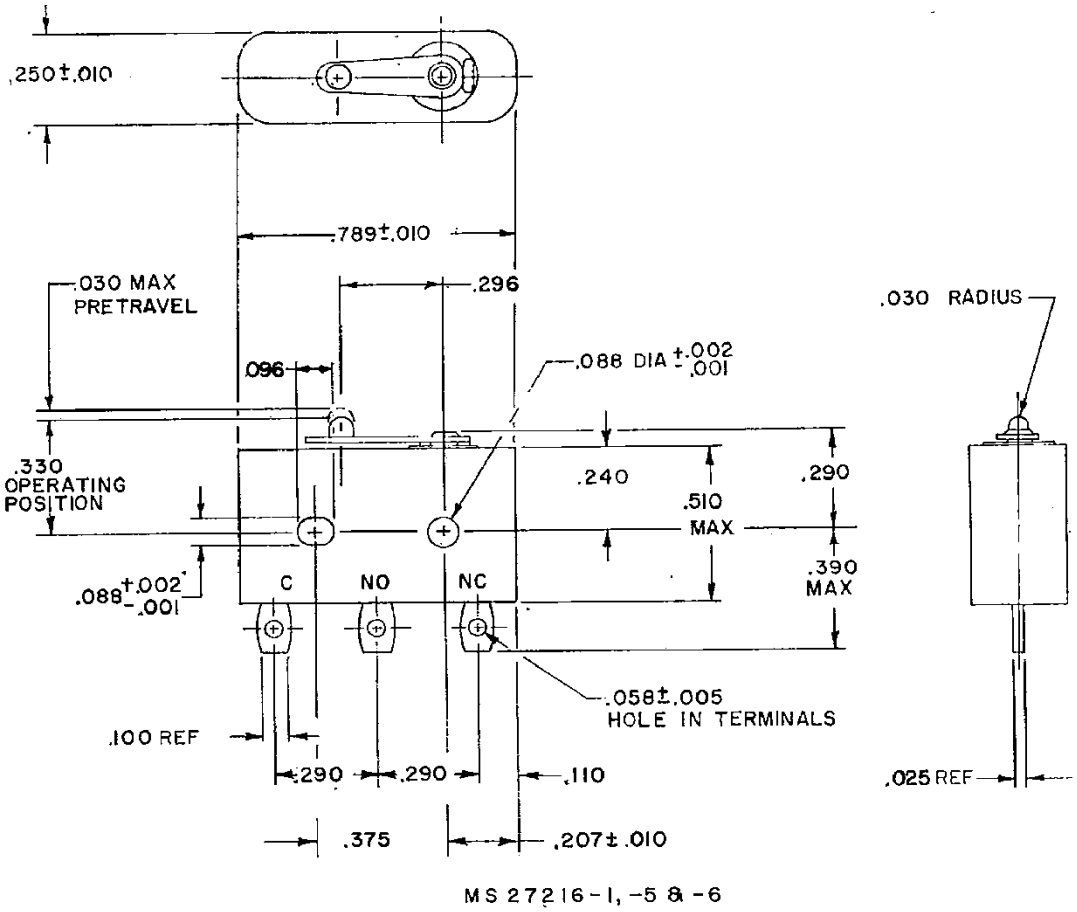


FIGURE 1. Dimensions and configurations.

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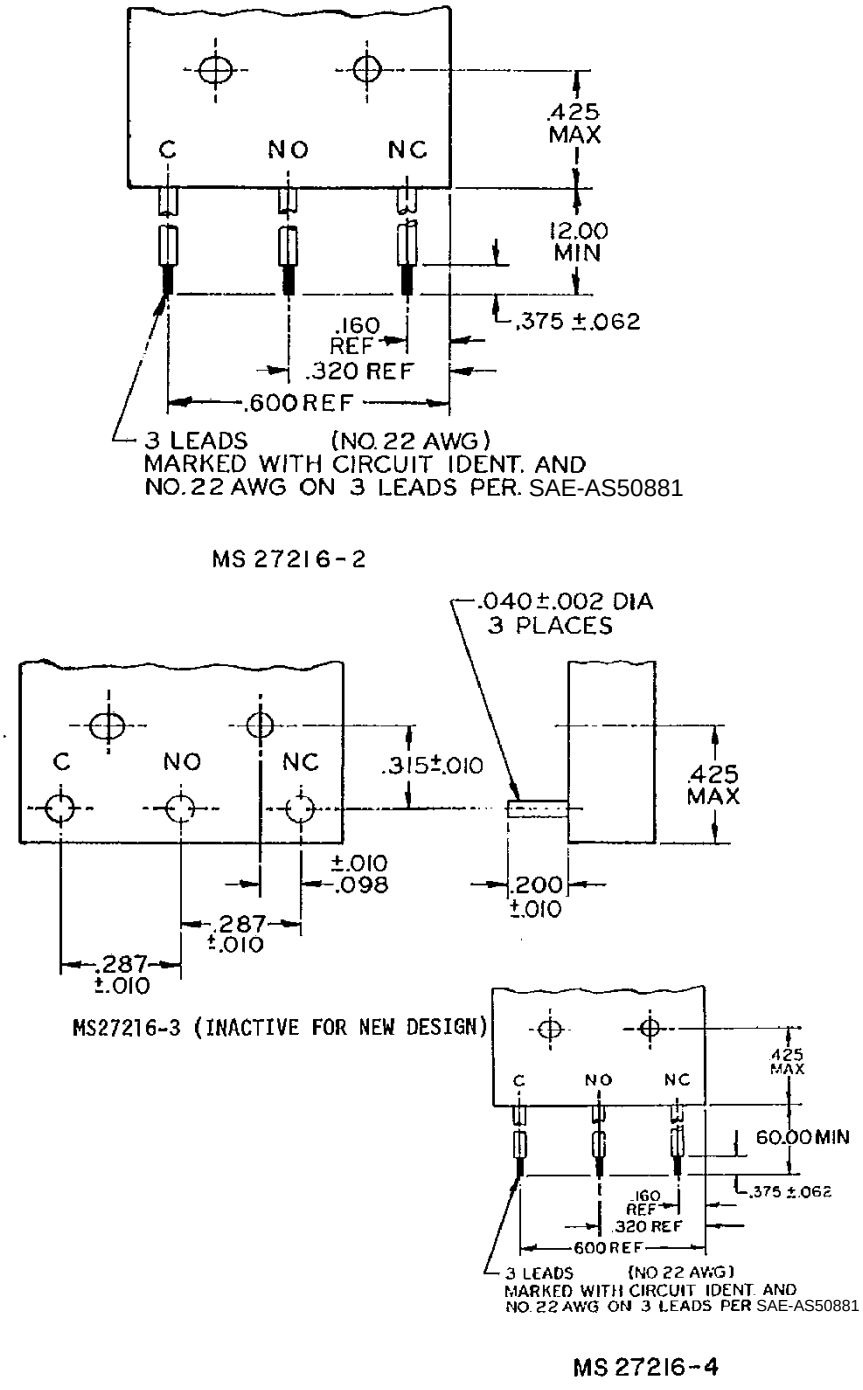


FIGURE 1. Dimensions and configurations -Continued.

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Inches	mm	Inches	mm	Inches	mm
.001	.03	.098	2.49	.315	8.00
.002	.05	.100	2.54	.320	8.13
.005	.13	.110	2.79	.330	8.38
.010	.25	.160	4.06	.375	9.52
.025	.64	.200	5.08	.390	9.91
.030	.76	.207	5.26	.425	10.80
.040	1.02	.240	6.10	.510	12.95
.058	1.47	.250	6.35	.600	15.24
.062	1.57	.287	7.29	.789	20.04
.088	2.24	.290	7.37	12.00	304.8
.096	2.44	.296	7.52	60.00	1524.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise stated, tolerance is ±.015 (0.38 mm).
4. Exact shape of switch optional provided dimensions specified are not exceeded.
5. The terminal designation shall be permanently and legibly marked on switch enclosure.
6. Mounting holes will accept pins or screws of .087 (2.21 mm) maximum diameter on .375 ±.002 (9.53 ±.05 mm) centers.
7. Military Standard Drawing MS27216 is canceled and superceded by MIL-PRF-8805/8.

FIGURE 1. Dimensions and configurations -Continued

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 5 (Hermetic)).

Temperature characteristic: 2 (-65° to +125°C).

Shock type:
MS27216-1, -2, -3, and –4: M(100 g's).
MS27216-5, and -6: H (high impact).

Vibration grade: 2 (10 to 2000 Hz).

Weight:
MS27216-1, -5 and –6: .014 pound maximum.
MS27216-2: .026 pound maximum.
MS27216-3: .020 pound maximum.
MS27216-4: .090 pound maximum.

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Operating characteristics:
Actuating force: 7 ounces maximum.
Release force: 1 ounce minimum.
Movement differential: .006 inch maximum.
Pretravel: .030 inch maximum.
Overtravel: .003 inch minimum.

Terminal strength: See table I.

Dielectric withstanding voltage:
At atmosheric pressure:
MS27216-1, -2, -3, -4, and –5: 1000 V rms.
MS27216-6: 800 V rms.
At reduced barometric pressure: 400 V rms
Following electrical endurance: In qualification inspection table, the dielectric withstanding voltage points of application between all unconnected terminals of the same pole shall be tested at 200 V rms minimum.

Contact resistance:
MS27216-1 through –5: Not applicable.
MS27216-6: 50 milliohms maximum initial, 100 milliohms maximum after mechanical endurance.

Mechanical endurance: 25,000 cycles.

Electrical endurance: 25,000 cycles.

Electrical ratings: See table II.

Low level circuit:

Applicable to MS27216-6: 25,000 cycles.

QUALIFICATION:

Group submission:
Sample: The number of sample units for each PIN covered by this specification sheet shall be as specified in table III.

Test routine: All sample units shall be subjected to visual and mechanical examination.
The sample units shall then be divided as specified in table III and subjected to the inspections indicated.

TABLE I. Terminal strength requirements.

Part or Identifying Number (PIN)	Applied force
MS27216-1, -5, -6	9 lbs.
MS27216-2, -3, -4	10 lbs.

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TABLE II. Electrical ratings.

Load	Sea level		70,000 feet 28 V dc
	28 V dc	115 V ac, 400 Hz	
	amperes	amperes	amperes
MS27216-1, -2, -3, -4			
Resistive Inductive Lamp	4 2 1	2 2 .5	4 2 1
MS27216-5			
Resistive Inductive	3 1	1 1	3 1
MS27216-6			
Resistive Inductive	0.5 0.25	- - - - - -	0.5 0.25

TABLE III. Qualification inspection.

Switches for testing	Number of sample units	Examination or test	Extent of approval
MS27216-1	30	Complete qualification table, as applicable, of MIL-PRF-8805	MS27216-1 MS27216-2 MS27216-4 MS27216-5 MS27216-6
MS27216-2	2	Insulation resistance Operating characteristics Terminal strength Vibration Shock Operating characteristics Visual and mechanical examination Seal	
MS27216-4	2	Visual and mechanical examination	
MS27216-5	24	Complete qualification table, as applicable, of MIL-PRF-8805 except for terminal strength, solderability, and explosion	
MS27216-6	6	Visual and mechanical examination Seal Dielectric withstanding voltage Insulation resistance Operating characteristics Overload Electrical endurance Low level circuit Dielectric withstanding voltage Operating characteristics	

Group C inspection:
Explosion test: Not applicable.

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Referenced documents. In addition to MIL-PRF-8805, this document references the following:

SAE-AS50881

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where modifications from this revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

Project (5930-2011-135)

Review activities:
Army – AR, AV, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/

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